Black-Hole Birth: Deeper Understanding

Not only did Wheeler and Oppenheimer differ profoundly on issues of national security, they also differed profoundly in their approach to theoretical physics. Where Oppenheimer hewed narrowly to the predictions of well-established physical law, Wheeler was driven by a deep yearning to know what lies beyond well-established law. He was continually reaching, mentally, toward the domain where known laws break down and new laws come into play. He tried to leapfrog his way into the twenty-first century, to catch a glimpse of what the laws of physics might be like beyond twentieth-century frontiers.

Of all the places that such a glimpse might be had, none looked more promising to Wheeler, from the 1950s onward, than the interface between general relativity (the domain of the large) and quantum mechanics (the domain of the small). General relativity and quantum mechanics did not mesh with each other in a logically consistent way. They were like the rows and columns of a crossword puzzle early in one's attempts to solve it. One has a tentative set of words written along the rows and a tentative set written down the columns, and one discovers a logical inconsistency at some of the intersections of rows and columns: Where the row word GENERAL demands an E, the column word QUANTUM demands a U; where the row word RELATIVITY demands an E, the column word QUANTUM demands a T. Looking at the row and column, it is obvious that one or the other or both must be changed to get consistency. Similarly, looking at the laws of general relativity and the laws of quantum mechanics, it was obvious that one or the other or both must be changed to make them mesh logically. If such a mesh could be achieved, the resulting union of general relativity and quantum mechanics would produce a powerful new set of laws that physicists were calling *quantum gravity.* However, physicists' understanding of how to marry general relativity with quantum mechanics was so primitive in the 1950s that, despite great effort, nobody was making much progress.